Do Now · retrieval

Paper 1 targeted reteaching

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does CIR hold?

Q2

What is a foreign key?

Q3

What is DNS used for?

Q4

Simplify A∨(¬A∧B), showing working.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Identify exact lost mark or incorrect idea

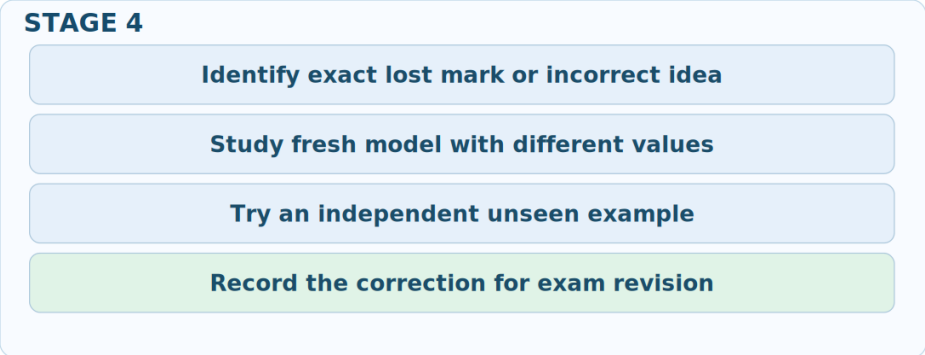
Core idea 2

Model — Study fresh model with different values

Core idea 3

Apply — Try an independent unseen example

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Evaluate — Record the correction for exam revision

Vocabulary · use the correct technical terms

reteach — Address an identified knowledge gap.

transfer — Apply a method to changed values.

method mark — Credit for correct intermediate reasoning.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A sequential fetch starts PC=120. Explain what MAR, MDR and CIR do.

Follow the worked method

1. Explain the method — MAR holds address, MDR transfers contents, and CIR holds instruction for decoding.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the method
Use exactly eight bits; inversion alone represents the wrong number.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

In a simple sequential fetch-decode-execute cycle, PC initially holds 70 and memory location 70 contains the instruction LOAD 25. State MAR, MDR, CIR and the next PC value, and explain one action of the control unit. Assume no branch or interrupt. [6]

Question 2

Convert −18 to eight-bit two's complement and explain sign. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Student(ID,Name) contains (1,Ada),(2,Ben). Mark(ID,Score) contains (2,80). Write an INNER JOIN using Student.ID=Mark.ID to select Name and Score, and state the output row. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Copying only the correct result proves a misconception has been resolved.

Use key vocabulary from memory

1. Explain reteach in your own words.

2. Explain transfer in your own words.

3. Explain method mark in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.